

# HR Overtime Management

## Odoo 19 Custom Module — Technical & Functional Documentation

Modules: **hr\_overtime\_management** (core) + **hr\_overtime\_payroll** (optional glue)

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## 1. Overview

This solution lets employees submit overtime requests linked to projects and tasks. Each request follows a multi-level approval chain (department manager → optional upper manager → HR), mirroring the pattern used by Odoo's *hr\_holidays* module. On final HR approval, the system can optionally create a timesheet line (*account.analytic.line*) and, if the payroll glue module is installed, push the overtime cost to a payslip input.

**Design principles:** Reusable approval engine (not overtime-specific), configurable pay-rate multipliers (never hardcoded), no parallel time-tracking table, and payroll integration isolated in a separate optional module.

## 2. Module Structure

| Module                 | Purpose                                                | Depends on                                |
|------------------------|--------------------------------------------------------|-------------------------------------------|
| hr_overtime_management | Core overtime requests, approval workflow, reporting   | hr, hr_timesheet, project, mail, resource |
| hr_overtime_payroll    | Contract-based hourly cost + payslip input on approval | hr_overtime_management, hr_payroll        |

## 3. Bug Fix Applied (Submit Error)

**Error:** *AttributeError: 'res.groups' object has no attribute 'users'* when clicking Submit on an overtime request.

**Cause:** In Odoo 19, the field on *res.groups* is named **user\_ids** (explicit members) and **all\_user\_ids** (including implied groups). The code incorrectly referenced *.users*, which does not exist.

**Fix:** In *hr.approval.chain.service.get\_hr\_responsible()*, changed *hr\_group.users* → *hr\_group.all\_user\_ids* so HR officers from the overtime HR group (and its implied groups) are resolved correctly.

## 4. Date & Time Change

Previously, overtime used a separate **date** field plus **start\_time** / **end\_time** as float fields with a time widget. These have been replaced with proper datetime pickers:

| Old field         | New field      | Type     | Description                         |
|-------------------|----------------|----------|-------------------------------------|
| date + start_time | start_datetime | Datetime | Start date & time picker (required) |
| date + end_time   | end_datetime   | Datetime | End date & time picker (required)   |

| Old field     | New field      | Type                    | Description                                          |
|---------------|----------------|-------------------------|------------------------------------------------------|
| date (manual) | date           | Date (computed, stored) | Auto-derived from start_datetime for filters/reports |
| —             | overtime_hours | Float (computed)        | Hours = (end_datetime – start_datetime) in hours     |

Overnight overtime is handled naturally: if end is on the next calendar day, the datetime difference still computes correctly (e.g. 22:00 → 02:00 next day = 4 hours).

## 5. Database Models & Columns

### 5.1 hr.overtime.type — Overtime Type (Configuration)

| Column          | Type                   | Description                                                                     |
|-----------------|------------------------|---------------------------------------------------------------------------------|
| name            | Char                   | Display name, e.g. Regular Overtime, Weekend, Public Holiday                    |
| code            | Char                   | Unique code per company (regular, weekend, holiday)                             |
| rate_multiplier | Float                  | Pay multiplier: Regular 1.5×, Weekend 2.0×, Holiday 2.5× (editable in Settings) |
| sequence        | Integer                | Display order in lists                                                          |
| company_id      | Many2one → res.company | Company scope (optional)                                                        |
| active          | Boolean                | Archive inactive types                                                          |

### 5.2 hr.overtime.request — Main Overtime Request

| Column              | Type                                 | Description                                                                 |
|---------------------|--------------------------------------|-----------------------------------------------------------------------------|
| name                | Char                                 | Auto sequence: OT/YYYY/MM/00001                                             |
| employee_id         | Many2one → hr.employee               | Employee who worked overtime (required)                                     |
| department_id       | Many2one → hr.department             | Related from employee, stored                                               |
| manager_id          | Many2one → hr.employee               | Direct manager (employee.parent_id), stored                                 |
| start_datetime      | Datetime                             | When overtime started (datetime picker)                                     |
| end_datetime        | Datetime                             | When overtime ended (datetime picker)                                       |
| date                | Date                                 | Computed from start_datetime; used in search filters and pivot              |
| overtime_hours      | Float                                | Computed duration in hours                                                  |
| overtime_type_id    | Many2one → hr.overtime.type          | Rate category (Regular/Weekend/Holiday)                                     |
| project_id          | Many2one → project.project           | Project for timesheet allocation                                            |
| task_id             | Many2one → project.task              | Task within the project                                                     |
| description         | Text                                 | Reason / details for the overtime                                           |
| attachment_ids      | Many2many → ir.attachment            | Supporting documents                                                        |
| state               | Selection                            | draft → submitted → manager_approved → upper_manager_approved → hr_approved |
| approval_line_ids   | One2many → hr.overtime.approval_line | Full approval audit trail                                                   |
| current_approver_id | Many2one → res.users                 | Who must act now (for record rules & My Approvals)                          |

| Column              | Type                             | Description                                                                        |
|---------------------|----------------------------------|------------------------------------------------------------------------------------|
| hourly_cost         | Monetary                         | Base hourly rate (0 in core; computed from contract in payroll module)             |
| total_cost          | Monetary                         | $\text{overtime\_hours} \times \text{hourly\_cost} \times \text{rate\_multiplier}$ |
| currency_id         | Many2one → res.currency          | Company currency                                                                   |
| company_id          | Many2one → res.company           | Company                                                                            |
| analytic_line_id    | Many2one → account.analytic.line | Timesheet line created on final approval (if enabled)                              |
| daily_hours_warning | Char                             | Warning if hours exceed company daily cap                                          |

### 5.3 hr.overtime.approval.line — Approval Trail

| Column        | Type                           | Description                                          |
|---------------|--------------------------------|------------------------------------------------------|
| request_id    | Many2one → hr.overtime.request | Parent request                                       |
| sequence      | Integer                        | Order in the chain (10, 20, 30...)                   |
| role          | Selection                      | dept_manager   upper_manager   hr                    |
| approver_id   | Many2one → res.users           | Assigned approver (HR: any officer in group may act) |
| state         | Selection                      | pending → to_approve → approved   refused            |
| decision_date | Datetime                       | When the decision was made                           |
| comment       | Text                           | Refusal reason or notes                              |

### 5.4 Reusable Engine — hr.approval.chain.service (Abstract)

Not stored in DB. Provides chain resolution for any future request type (expenses, permissions, etc.):

- `resolve_chain(employee, chain_builder, hr_group_xmlid)` — generic entry point
- `build_manager_hr_chain(employee)` — standard dept manager → upper manager → HR
- `get_hr_responsible(employee)` — picks HR user from `group_overtime_hr_officer`
- `_sanitize_chain(chain)` — removes duplicates and empty entries

### 5.5 hr.approval.chain.mixin (Abstract)

Mixed into `hr.overtime.request`. Provides workflow methods:

- `action_submit` → resolve chain, create approval lines, schedule activity
- `action_approve_current` → validate approver, advance state, trigger next activity
- `action_process_refusal` → stop chain, set state=refused
- `_on_approval_complete()` hook → create analytic line / payslip input

### 5.6 res.company Extensions

| Column                                       | Type     | Description                                          |
|----------------------------------------------|----------|------------------------------------------------------|
| <code>overtime_generate_analytic_line</code> | Boolean  | Auto-create timesheet on HR approval (default: True) |
| <code>overtime_default_type_id</code>        | Many2one | Default overtime type for new requests               |

| Column                   | Type  | Description                          |
|--------------------------|-------|--------------------------------------|
| overtime_daily_hours_cap | Float | Warning threshold (default: 4 hours) |

### 5.7 hr.employee Extensions

| Column                 | Type     | Description                          |
|------------------------|----------|--------------------------------------|
| overtime_request_count | Integer  | Count of approved overtime requests  |
| overtime_hours_ytd     | Float    | Approved overtime hours year-to-date |
| overtime_cost_ytd      | Monetary | Approved overtime cost year-to-date  |

## 6. Approval Workflow

**Chain resolution** (same pattern as hr\_holidays):

- Step 1: employee.parent\_id (or department manager if no parent)
- Step 2: parent of dept manager, if different person
- Step 3: HR officer (always mandatory final stage)

**Two scenarios:**

- **2 stages:** Dept manager only (no one above) → dept\_manager → HR
- **3 stages:** Dept manager + upper manager → dept\_manager → upper\_manager → HR

**States after each approval:** submitted → manager\_approved → (upper\_manager\_approved if 3-stage) → hr\_approved. Refusal at any stage immediately sets state=refused.

## 7. Security Groups & Record Rules

| Group                        | Access                                                       |
|------------------------------|--------------------------------------------------------------|
| group_overtime_user          | Create/read own requests; write only while draft             |
| group_overtime_hr_officer    | Read all; write at HR-pending stage; final approval          |
| group_overtime_admin         | Configure types, settings; full access                       |
| Managers (no separate group) | Read/write requests where current_approver_id = current user |

## 8. Cost Calculation

**Formula:** total\_cost = overtime\_hours × hourly\_cost × overtime\_type.rate\_multiplier

**Core module:** hourly\_cost = 0 (no payroll dependency)

**hr\_overtime\_payroll:** hourly\_cost = contract.wage ÷ company.overtime\_hours\_per\_month (default 173.33)

## 9. Configuration (Settings → Employees → Overtime)

- Generate timesheet on approval
- Default overtime type
- Daily hours warning cap
- Link to payroll (installs hr\_overtime\_payroll module)

## 10. Menus & Views

- Overtime → My Requests — employee self-service
- Overtime → My Approvals — kanban for pending approvers
- Overtime → All Requests — HR officers see everything

- Overtime → Configuration → Overtime Types — admin only
- Views: list, form (statusbar + approval tab + chatter), kanban, pivot, graph, PDF report

## 11. Upgrade Instructions

After pulling these changes, upgrade the module:

- Apps → HR Overtime Management → Upgrade
- Or CLI: `python odoo-bin -c debian/odoo.conf -d YOUR_DB -u hr_overtime_management --stop-after-init`

Existing records with old float time fields are migrated automatically via `migrations/19.0.1.0.1/post-migrate.py`.

## 12. Assign HR Officers

For Submit to work, at least one user must belong to the **Officer: Overtime HR Approval** group (Settings → Users → HR Overtime HR Officer). Without this, the approval chain cannot resolve the final HR stage.